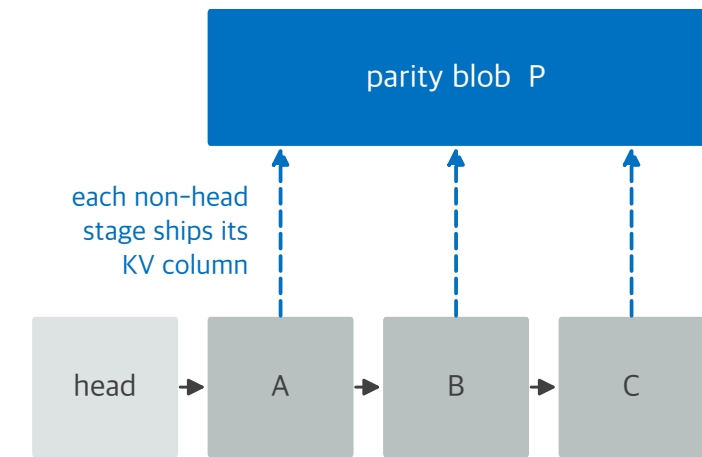


# ① Steady state

one blob, not one copy per stage

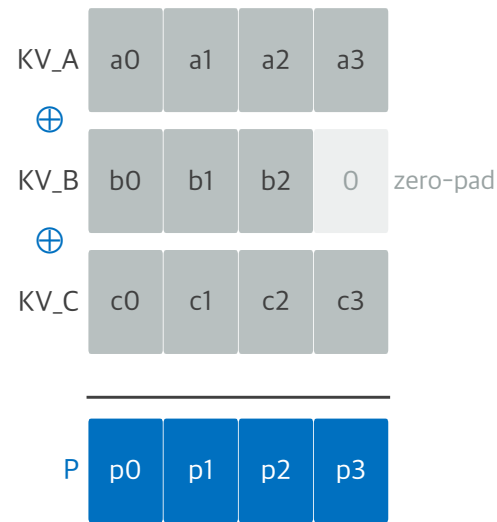


head is coordinator-sourced —  
not in the parity group

$$P = KV\_A \oplus KV\_B \oplus KV\_C$$

# ② Byte-wise XOR

raw bytes — works for any dtype

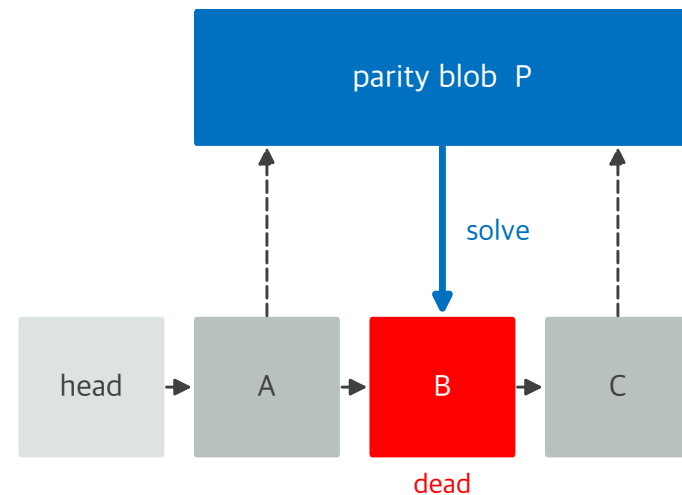


one column per position, per stage

# ③ Recovery

zero forward passes

survivors give back what they still hold



$$KV\_B = KV\_A \oplus KV\_C \oplus P$$